

MFE 230GA · FINAL PROJECT · FALL 2026

Follow the Workers

Do labor-market flows predict which U.S. industries outperform?

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1 Executive summary

Can public labor-market flows predict which of 13 U.S. industry groups outperform? We rank the groups each month on the year-over-year change in openings, hires, quits, layoffs, hours and wages, as known at the time, and hold a market-neutral book long the top three and short the bottom three. The design was preregistered, frozen and opened once on 142 untouched months (November 2014 to August 2026). The primary strategy lost 4.6% a year after costs (Sharpe -0.58); its forecasts were uncorrelated with returns and indistinguishable from a placebo. We conclude: *Do not implement*. It failed because the preregistered signs of layoffs and hours were wrong, the one stable one-month signal (wage growth) was dropped during the build, and the labor market never tightened in the research years, so tightness went untested. The data are consistent with a slower, opposite effect, as investment-based asset pricing predicts: industries that hire and lose workers fastest earn lower returns over the following 9 to 24 months. A 12-month book on that sign is post hoc, lost money in the first test half and does not survive deflation, though positive in every window (test Sharpe $+0.29$); it runs as a paper book in a forward test frozen on 3 October 2026. Claude was a useful auditor and idea generator when a human checked every claim and it could not change the rules after seeing results.

2 What did we try? Labor flows as industry news

2.1 Idea: demand news, cost news and tightness

Our thesis is that the labor market reports an industry's prospects before its earnings do. The Bureau of Labor Statistics (BLS) publishes job openings, hires, quits and layoffs by industry every month in the Job Openings and Labor Turnover Survey (JOLTS), and hours and earnings in the Current Employment Statistics (CES). The series are free and timely, while published industry-rotation strategies rely on price momentum (Moskowitz and Grinblatt, 1999; Zarattini and Antonacci, 2024). We state three hypotheses (Figure 1).

- *H1, demand channel* (preregistered and tested). Rising openings, hires and hours signal growing demand for an industry's output before it reaches reported sales, so the industry outperforms over the next one to three months (Hong et al., 2007).
- *H2, cost and investment channel* (tested post hoc). Firms that hire heavily convert growth options into assets in place and pay adjustment costs, so their expected returns fall over the following 6 to 24 months (Belo et al., 2014); wage pressure squeezes margins, and labor-market frictions make labor-intensive firms riskier (Kuehn et al., 2017; Donangelo et al., 2019).
- *H3, tightness moderator*. When vacancies outnumber the unemployed, hiring is expensive and H2 should dominate H1. We measure tightness by $T_t = \log(\text{job openings}_t / \text{unemployed}_t)$; $T > 0$ means more vacancies than unemployed ($V/U > 1$).

The preregistered strategy traded H1 at a one-month horizon. H2's horizon range was written down on 3 October 2026, after the test had been opened and after the horizon profile in Figure 8 had been seen; it carries the post-hoc label wherever it appears.

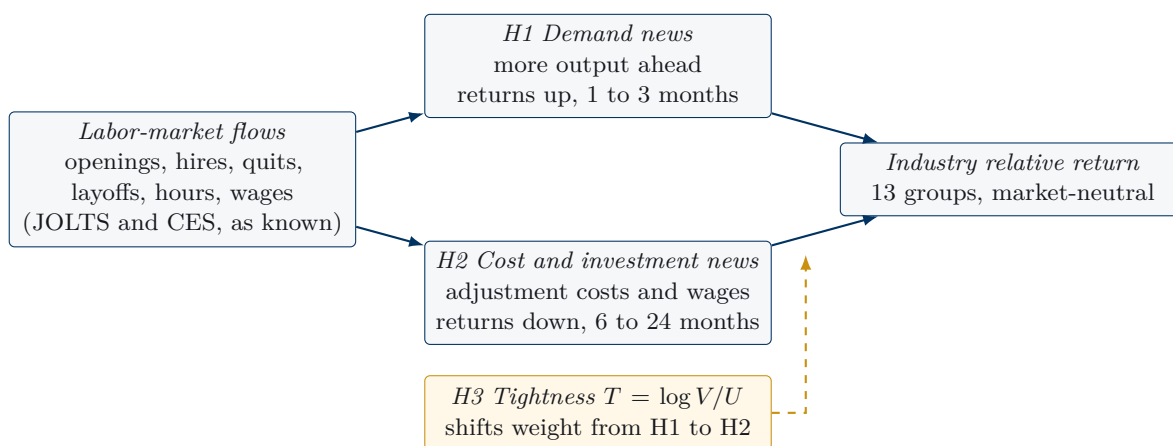


FIGURE 1. The three hypotheses. Labor flows carry two kinds of industry news with opposite sign and different horizons (H1, H2); tightness should change which one dominates (H3).

Table 1 lists each signal, the sign we preregistered under H1 and the mechanism behind it. The last column, added after the fact, shows what the data say.

TABLE 1. Signals, preregistered signs and what the data say. CONFIRMATORY (design) / POST-HOC (last column)

Signal	Definition (group g , month t)	Prior	Mechanism	IC, $h=1$
F1 openings	12-month change in 3-month average openings rate	+	H1 demand news	−0.005
F2 hires	same, hires rate	+	H1 demand news; H2 at long horizons	+0.013
F3 quits	same, quits rate	none	H2, worker bargaining power	+0.009
F4 layoffs	same, layoffs rate	−	H1, negative demand shock	+0.028
F5 hours	12-month log change, average weekly hours	+	H1, intensive margin	−0.010
W wages	12-month log change, average hourly earnings	dropped	H2 cost pressure (−) vs. labor risk premium (+)	+0.039

IC (information coefficient) is the average monthly Spearman correlation between the signal and the next month's industry relative return, May 2004 to August 2026, revised data with publication lags. Throughout, t -statistics use Newey–West standard errors, which allow for autocorrelation (HAC); only W has $|t| > 2$ ($t = 2.50$). The preregistered fixed rule A0 used $z(F1) + z(F2) - z(F4) + z(F5)$; the data reject the signs of F4 and F5.

2.2 Data: point-in-time labor flows and industry returns

TABLE 2. Data sources (all free; links in Appendix B).

CONFIRMATORY

Source	Series	Freq.	Known at t	Vintage handling in the frozen study
BLS JOLTS	openings, hires, quits, layoffs rates; 13 industries; not seasonally adjusted	M	$t - 2$	ALFRED vintages: the value known on each decision date
BLS CES	average weekly hours and hourly earnings, production and nonsupervisory workers	M	$t - 1$	as above
FRED	total openings (JTSJOL) and unemployed (UNEMPLOY) for T_t	M	$t - 2$	as above
Ken French	49 value-weighted industry portfolios $\rightarrow$ 13 groups; five factors and momentum	M	t	CRSP build 202608

Universe, a design constraint. The 49 French industries are mapped to the 13 JOLTS/CES industry groups and value-weighted within each group (Appendix C). With 13 groups the achievable information ratio is small: by the fundamental law of active management (Section 3.3) it is about $IC \times \sqrt{BR} \times TC$, where BR is breadth (independent bets a year) and TC the transfer coefficient (the correlation between the signal and the weights held); a monthly book on 13 groups has BR of at most $13 \times 12 = 156$ and a 12-month book 13, so a monthly IC of 0.05 caps the ratio near 0.6 and a 12-month IC of 0.10 near 0.35. The mapping is also coarse: labor data describe every establishment in an industry, returns describe listed firms, and durable manufacturing is 60% semiconductors by market capitalisation.

Point-in-time data. Labor data are revised after first release, so today's values leak information. The frozen study rebuilt every series as it stood on each decision date from ALFRED, the St. Louis Fed archive of data vintages (78 series), and kept three alternatives: first release, fully revised, and revised with fixed publication lags. The primary result barely depends on the choice: A3's test Sharpe is -0.58 as known, -0.52 on first releases, -0.56 revised and -0.53 with fixed lags. Our post-hoc work (Section 3.3) uses revised data with fixed lags, which reproduces the frozen fixed-lag signal with a correlation of 0.998 across all group-months; as-known data pick the same top-three groups in only 42% of months, so post-hoc holdings, though not the conclusions, would have differed in real time. One exception, found in the final code review: the federal shutdown of October–November 2025 delayed BLS releases, so two post-hoc earning months (November and December 2025) use labor data published after the decision date; the confirmatory as-known data are unaffected, and the v2/v3 numbers were left as frozen (docs/OPEN_ISSUES.md).

Sample. Research (signal design and model selection): May 2004 to September 2014, 125 months. Test, opened once: November 2014 to August 2026, 142 months. Figure 2 shows why H3 could not be learned: V/U never exceeded 1 in the research period and exceeded it in 75 test months.

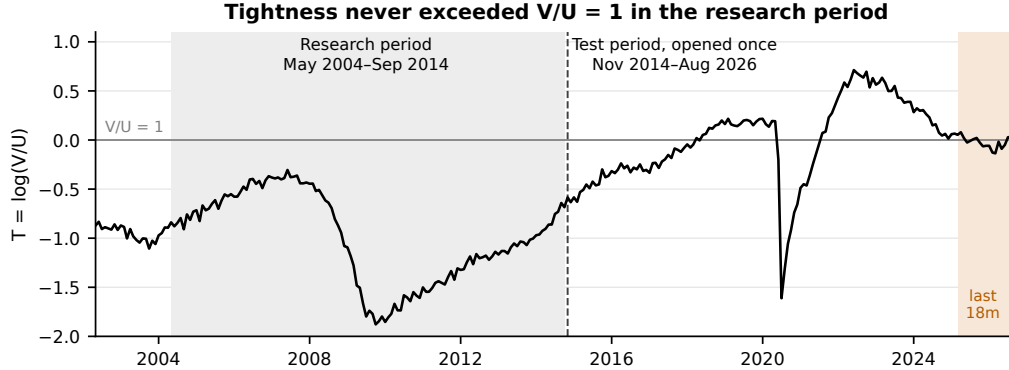


FIGURE 2. Tightness never exceeded $V/U = 1$ in the research period, so a moderator learned there (H3) is extrapolated throughout the test.

POST-HOC

2.3 Implementation: signals, models, portfolio and AI agents

Signals. For a labor series $x_{g,t}$ in group g , let $\bar{x}_{g,t} = \frac{1}{3}(x_{g,t} + x_{g,t-1} + x_{g,t-2})$. With publication lag ℓ (2 months for JOLTS, 1 for CES),

$$F_{g,t} = \bar{x}_{g,t-\ell} - \bar{x}_{g,t-\ell-12} \quad (\text{log differences for hours and earnings}). \quad (1)$$

Each signal is divided by its own expanding-window standard deviation (at least 24 months), z-scored across the 13 groups and winsorised at ± 3 , giving $z_{g,t}^{(k)}$. Year-over-year differences remove seasonality, so the not-seasonally-adjusted series that exist for every industry can be used.

Models. Table 3 lists the five preregistered strategies, all of them H1 books. The learned ones are ridge regressions of next-month relative return on the signals,

$$\hat{\beta}_t = \arg \min_{\beta} \sum_{s < t} \sum_g (r_{g,s+1}^{\text{rel}} - z_{g,s}^{\top} \beta)^2 + \alpha \|\beta\|^2, \quad (2)$$

re-estimated each month on an expanding window, with α chosen once by cross-validation on contiguous time blocks with a one-month purge, inside the research period, over a grid from 0.01 to 1000. Cross-validation chose 1000, the strongest shrinkage the grid allowed, for every model.

TABLE 3. Preregistered strategies (legend for the codes used below).

CONFIRMATORY

A0	Fixed rule, no estimation: $S_{g,t} = \frac{1}{4}(z^{(1)} + z^{(2)} - z^{(4)} + z^{(5)})$.
A1	Ridge on F1–F5.
A1-T	A1 plus each signal interacted with tightness T (H3).
A3	A1 plus features proposed by <i>independent</i> LLM agents (Section 2.3); the primary, chosen on research data.
A4	A1 plus features proposed by <i>communicating</i> LLM agents.

Portfolio, risk model and costs. Each month the score $s_{m,g}$ known at the end of month $m - 1$ is converted to a dollar-neutral tranche (one month's positions), long the three highest-ranked groups and short the three lowest; the book averages the last three tranches, is hedged to zero market beta, scaled to a 10% ex-ante volatility using a Ledoit–Wolf shrinkage covariance (Ledoit and Wolf, 2004) and capped at a gross exposure of 2, and pays 10 bp per unit of industry turnover plus 2 bp on the hedge:

$$\tau_{m,g} = \begin{cases} +\frac{1}{3} & g \in \text{top 3 of } s_{m,\cdot} \\ -\frac{1}{3} & g \in \text{bottom 3 of } s_{m,\cdot} \\ 0 & \text{otherwise} \end{cases}, \quad b_m = \frac{1}{H} \sum_{j=0}^{H-1} \tau_{m-j}, \quad H = 3, \quad (3)$$

$$h_m = - \sum_g b_{m,g} \hat{\beta}_{m,g}, \quad x_m = (b_{m,1}, \dots, b_{m,13}, h_m), \quad (4)$$

$$k_m = \min \left(\frac{0.10}{\sqrt{12 x_m^{\top} \hat{\Sigma}_m x_m}}, \frac{2}{\|x_m\|_1} \right), \quad w_m = k_m x_m, \quad (5)$$

$$r_m^{\text{net}} = \sum_g w_{m,g} r_{m,g}^e + w_{m,M} \text{MKT}_m - c \sum_g |w_{m,g} - \tilde{w}_{m,g}| - c_h |w_{m,M} - \tilde{w}_{m,M}|. \quad (6)$$

where $s_{m,g}$ is the score of group g known at the end of month $m - 1$, $\hat{\beta}_{m,g}$ and the Ledoit–Wolf covariance $\hat{\Sigma}_m$ (13 groups plus the market) use the 60 months ending in $m - 2$ (at least 36), $r_{m,g}^e$ is the group's excess return, $\tilde{w}$ are the previous weights after drift, $c = 10$ bp and $c_h = 2$ bp. In the test the cap, rather than the volatility target, bound in 87% of A3's months, so realised volatility ran at 7.8% against the 10% target.

HOW WE KEPT THE TEST HONEST

- Every input as it was known on the decision date (ALFRED vintages); the primary result is unchanged under four data vintages.
- Signals, models, the primary strategy and five pass/fail gates (G1 timing, G2 prediction, G3 incremental alpha, G4 robustness, G5 implementability) were fixed and hashed before any test data were touched; the test period was opened exactly once.
- Every result reported whatever its sign: a circular-shift placebo (the signal panel is rotated against returns by every shift from 24 months up, with the model refitted each time, and the real IC is compared with the shifted ones), drop-one-industry checks, bootstrap comparisons of the four registered strategy pairs with a Holm correction for multiple tests, and a deflated Sharpe ratio (DSR), which discounts a

Sharpe ratio for the number of trials, or “looks”, taken (Bailey and López de Prado, 2014).

- Anything decided after seeing test results is labelled **POST-HOC** and was itself written to a file and hashed before it ran; the paper book is labelled **FORWARD TEST**. [TODO: Alex: confirm the freeze hashes and the single opening against FREEZE.json and evaluation_started.json]

AI-assisted signal search. We asked large language model (LLM) agents to propose new labor features from six anonymised variables using a restricted formula grammar. Two arms of three agents each, *independent* versus *islands* that exchange short migration reports, ran three repetitions, for 108 proposals per study (Figure 3). Selection was mechanical (research $t \geq 1.5$, same sign in both halves, low correlation, at most three per arm). An LLM judge, audited by a human, scored whether ideas spread between agents. The agents were Claude Sonnet 5.5 in the confirmatory study and Claude Opus 5.5 in an exploratory follow-up run after the test had been seen (Appendix F).

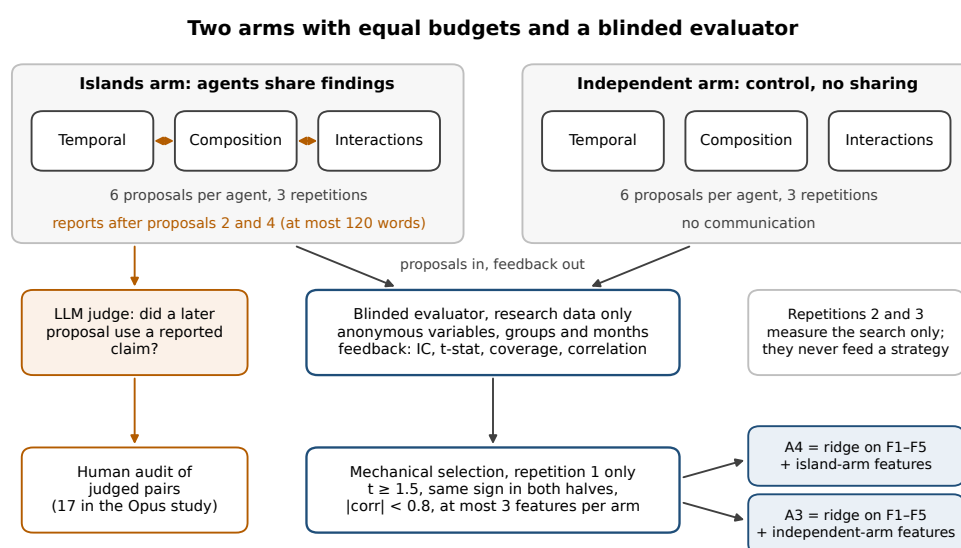


FIGURE 3. Two arms with equal budgets, a blinded evaluator, mechanical selection and a human-audited LLM judge.

Claude as research assistant. We used Claude (Anthropic) at four points of the build: auditing sources and code (P1), proposing signals as blinded agents (P2), judging whether ideas spread (P3) and red-teaming the robustness plan (P4). We used it again to review the finished study (P5) and to run this iteration (P6). Three of the prompts, verbatim, are below; outputs and our evaluation are in Section 3.4, transcripts in Appendix A.1.

PROMPT P1: SOURCE AND CODE REVIEW

Find mapping errors, timestamp or look-ahead errors, and code bugs in the material below. For each problem, give the exact line or row, why it is wrong, and a test that would confirm it. Do not report style issues. [followed by the series manifest, the industry crosswalk and the as-of code]

PROMPT P2: BLINDED SIGNAL-SEARCH AGENT (SYSTEM PROMPT, OPENING)

You are a quantitative researcher. You will propose one feature per turn to predict which of 13 anonymous groups (G01 to G13) will have higher returns next month relative to the others. You see only the variable dictionary below, the allowed operations, and summary statistics of your previous submissions. You do not know what the groups, variables or time periods are. Do not guess them and do not use knowledge about any real industry, company or date. [variable dictionary, grammar and JSON format follow]

PROMPT P4: ROBUSTNESS RED TEAM

Give 10 distinct reasons these results could be spurious, ranked by likelihood. For each, name a test that can be run with this data and the result that would make us say "do not implement". [followed by the research-period result tables, without dates or names]

3 What did we learn? Slow cost news, not fast demand news

ANSWER TO THE THESIS

- *H1, demand channel: rejected CONFIRMATORY*. The primary H1 book lost 4.6% a year (Sharpe -0.58), its IC was -0.021 ($t = -1.08$), the placebo p was 0.72 and four of five gates failed.
- *H2, cost channel: consistent with the data, not established POST-HOC*. The hiring score has a 12-month IC of $+0.096$ ($t = 2.43$) and a 12-month book is positive in every window ($+0.29$ test), but it fails the pre-declared reading rule: it lost money in the first half of the test and its deflated Sharpe is 0.05 after 132 looks. A forward test is live.
- *H3, tightness: untestable as designed*. V/U never exceeded 1 in the research sample, so 133 of 142 test months fell in one tightness bin.

Table 4 reports the one-time test of H1; no strategy made money, and costs are not the explanation, since A3 paid 1.1% a year in trading costs against a gross loss of 3.5%.

TABLE 4. Preregistered H1 strategies: research versus the one-time test.

CONFIRMATORY

Strategy	Research	Test, Nov 2014–Aug 2026 (142 months)			
	Sharpe	Sharpe	Net return	Volatility	Max drawdown
A0 fixed rule	-0.44	-0.36	-3.5%	9.7%	-40.3%
A1 ridge	-0.54	-0.51	-4.0%	7.9%	-37.7%
A1-T tightness	-0.55	-0.59	-4.9%	8.2%	-36.3%
A3 primary	-0.13	-0.58	-4.6%	7.8%	-41.1%
A4 islands	-0.18	-0.79	-6.1%	7.8%	-53.3%

Annualised, net of costs, as-known data; research Sharpe over the 88 months in which every model had a forecast. Primary: factor alpha -0.38% a month ($t = -1.86$), Sharpe -0.79 at 25 bp costs, 41% maximum drawdown; every Holm-adjusted pairwise comparison had adjusted $p = 1.0$.

3.1 Style and risk exposures: partly a quality and growth tilt

The H1 books are quality and growth tilts rather than bets on labor news. Regressed on the Fama–French five factors, momentum and an industry-momentum factor (Figure 6, appendix), the primary loads on profitability (RMW $+0.19$, $t = 2.2$) and against value (HML -0.15 , $t = -2.0$), with an alpha of -0.38% a month ($t = -1.86$). The tilt is plausible: industries adding workers and hours are the profitable, growing ones, so a long-hiring book buys profitability and sells value. The post-hoc wage-growth book tilts the same way but harder (HML -0.43 , $t = -4.4$; momentum $+0.22$, $t = 3.3$), while the H2 book loads the other way, against profitability (RMW -0.36 , $t = -4.45$).

3.2 Performance over time: every preregistered book lost

The H1 primary lost money in every window except the last 18 months, and that bright spot ($+0.85$) rests on an IC t of 0.44 (Figure 4; Table 9 in Appendix D has every window). For A3, “post-2010” is in effect the test, which starts in November 2014; “full” adds the research years and is descriptive. The post-hoc books show the opposite pattern: wage growth did well until 2024 and lost heavily in the last 18 months (-1.71), while the post-hoc H2 book is positive in every course window, although

its two test halves split -0.21 and $+0.64$ (Table 6).

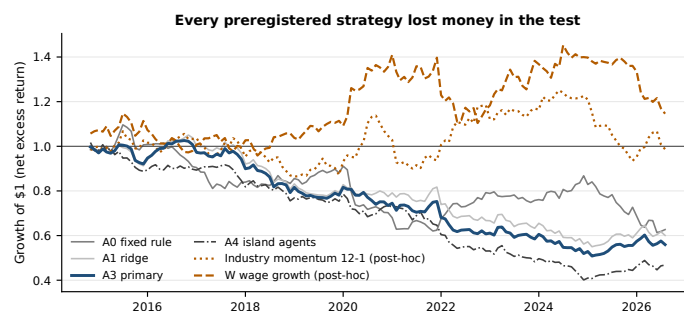


FIGURE 4. Every preregistered H1 strategy lost money in the test; the post-hoc wage-growth book gained until 2024 and gave much of it back.

CONFIRMATORY (A0–A4) / POST-HOC (momentum, W)

3.3 Robustness and limits: why H1 failed, what survives

What 13 industries can deliver. In the test the fundamental law implies IRs of -0.21 for A3 (IC -0.021 , TC 0.80), $+0.56$ for W and $+0.17$ for the H2 book (12-month IC $+0.096$, TC 0.48) against realised -0.58 , $+0.16$ and $+0.29$; the preregistered overlay (hedge, volatility target, binding cap, 0.9 turnover a month) cost the monthly books most of what their IC allowed.

Why H1 failed: four causes. Four post-hoc explanations (figures in Appendix D).

1. *Signs not supported by the data.* Layoffs carry a positive IC in both periods and hours a negative one, the opposite of the signs A0 imposed (Table 1); wage growth, dropped during the build, is the only signal whose IC exceeds 1.5 standard errors in both periods ($+0.029$ research, $+0.049$ test).
2. *Wrong horizon.* At one month no hiring signal predicts returns; at twelve months hires (-0.062 , $t = -2.1$) and quits (-0.065 , $t = -1.7$) carry the H2 sign.
3. *An untestable moderator (H3).* Tightness terciles were set on an expanding window, so 133 of 142 test months fell in the “high” tercile; a split on whether T rose or fell over 12 months (88/54 test months) would have been testable.
4. *Fragile agent features.* The two features selected for the primary were gated to the “low tightness” regime and active in 3 of 142 test months; every selected feature, in both studies, failed the frozen test criterion.

Losses concentrated in health care (26 points of cumulative return), construction (19) and finance (16), against 13 points of trading costs.

What survives: the 12-month cost channel. None of the four post-hoc variants meets the pre-declared reading rule (positive Sharpe at 25 bp, IC $t \geq 2$, alpha $t \geq 2$, positive IC in both test halves, placebo $p < 0.05$, DSR above 0.95). We froze them in a hashed file (`analysis/specs/study2_1_variants.md`) after the test was opened, ran each once and report all four (Table 5, Figure 9). The declared headline, a wage-growth rank (W), has the right IC ($+0.049$, $t = 2.04$, placebo $p = 0.02$) but no alpha ($t = 0.29$) and a DSR of 0.02; a composite of all six signals with signs learned on research data (SC) is negative, and the average rank of wage growth and 12-month industry momentum (W+MOM) reaches $+0.35$ because momentum supplies the return.

HQ12 is the H2 book: $-\frac{1}{2}(z(F2) + z(F3))$, short the industries that hire and lose workers fastest, held twelve months. Its score has no one-month IC (-0.020 , $t = -0.87$) and a positive IC from 9 to 24 months, peaking at 12 ($+0.096$, $t = 2.43$; Figure 5); its alpha is $+0.26\%$ a month ($t = 1.48$ test, 1.92 full sample, 12 Newey–West lags; Table 5 uses 6), and it fails the rule. Where HQ12 came from matters. Its sign is the one the investment channel predicts (Belo et al., 2014); its 12-month horizon was chosen from a horizon profile that included the test period (Figure 8), so the horizon is fitted on the test. On the research years alone the 12-month IC is $+0.047$ ($t = 0.90$) with a horizon-matched

placebo p of 0.26: the research sample alone would not have selected it.

TABLE 5. Post-hoc variants frozen in `analysis/v2_spec.md`: net Sharpe by window and test statistics at 10 bp. Every column, the test window included, is post hoc: the variants were chosen after the test had been opened. Test figures are the test slice of a continuous run from May 2004 (A3’s sealed ledger starts from an empty book in November 2014). IC, alpha t and placebo are one-month statistics with 6 Newey–West lags; DSR counts the 112 looks declared in that file as independent trials, a rough deflation since the looks are correlated.

POST-HOC

Variant	Research	Test	Full	Post-2010	Last 18m	Test IC (t)	Alpha t	Placebo p	DSR
W	+0.09	+0.16	+0.15	+0.12	−1.71	+0.049 (+2.04)	+0.29	0.02	0.02
SC	−0.06	−0.15	−0.11	−0.16	−1.34	+0.042 (+1.65)	−0.84	0.05	0.00
W+MOM	+0.27	+0.35	+0.32	+0.24	−0.78	+0.032 (+1.21)	+1.36	0.26	0.09
HQ12	+0.50	+0.29	+0.35	+0.30	+0.29	−0.020 (−0.87)	+1.37	0.85	0.06

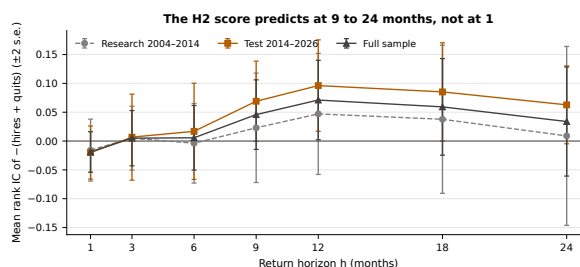


FIGURE 5. Post-hoc horizon profile of the H2 score: no one-month IC and a positive IC from 9 to 24 months, peaking at 12 (test +0.096, $t = 2.43$).

POST-HOC

How robust is the H2 book? Table 6 lists the seven pre-declared checks, frozen in `analysis/specs/study2_2_hq12_robustness.md` with the look count raised to 132 and run once. HQ12 fails the check on the test halves; after 132 looks its DSR is 0.05 in the test and 0.16 over the full sample, so we read it as a hypothesis consistent with the economics and short of evidence. Its behaviour across tightness regimes is recorded without a claim in Figure 12.

TABLE 6. HQ12 robustness battery, test window unless stated; the bootstrap condition was pre-declared as “test or full-sample interval excludes zero”.

POST-HOC

Check	Result	Holds?
Components, held 12 months	hires alone +0.19, quits alone +0.16	yes
Holding period	Sharpe +0.14 (9 months), +0.29 (12), +0.30 (18)	yes
Both test halves positive	first half −0.21, second half +0.64; 12-month IC positive in both (+0.097, +0.095)	no
Costs	break-even 68 bp; +0.22 at 25 bp, +0.09 at 50	yes
Drop one industry	test Sharpe between +0.18 (without health care) and +0.48 (without real estate)	yes
Block bootstrap (12-month blocks, 5,000 draws)	90% interval [−0.07, +0.65] test; [+0.05, +0.64] full sample	yes (full)
Placebo	one-month $p = 0.85$ (Table 5); horizon-matched $p = 0.06$ test, 0.04 full, on overlapping returns, added after the run	outside rule

Live forward test, Q4 2026. Four paper books with no capital behind them were frozen on 3 October 2026 (hash 735a8ffa, amended by c76cde86 after a data-release check stopped the first run) for the 30 September decision, from data published by 29 September; each has three longs, three shorts and a SPY beta hedge, held to 31 December (Appendix G). W+MOM and HQ12 were added after the historical v2 results had been seen, a selection step recorded as one. One quarter has little statistical power: a Q4 return between −10% and +10% is consistent with no skill, and no position of any size is recommended.

Limits and power. With 142 test months only a Sharpe near 0.6 reaches $t \approx 2$, so a null does not rule out a smaller effect; the best post-hoc test Sharpe, W+MOM's +0.35, corresponds to $t \approx 1.2$ over those months, and none of the post-hoc Sharpe ratios is distinguishable from zero after deflation. The 49-to-13 mapping dilutes the labor signal; the one near-positive result in the Opus follow-up came from a single long position in a group that is 60% semiconductors (Appendix F). All post-hoc numbers use revised data with fixed lags; the as-known rerun was not run for lack of an API key. Stock-level baskets by NAICS code from CRSP are the follow-up we would run next.

3.4 Claude evaluation: useful auditor, unreliable judge

Claude was the AI research assistant in every role, which this term's version of the course allows; the switch from the ChatGPT template is logged as preregistration amendments 001 and 002 (outputs/deviations.md in the frozen study). Table 7 records every interaction; Appendix A has the detail and Appendix F compares the Opus and Sonnet agents. The models are taken from the call logs: Sonnet 5.5 at medium reasoning effort for the 199 logged calls of the confirmatory study, Opus 5.5 at extra-high reasoning effort for the 294 calls of the follow-up. [TODO: Alex: exact name and vendor of the engineering assistant that wrote the pipeline code ("6 astra"); it is not documented in the repository]

When the judge and the human disagreed. In the follow-up we hand-checked 17 report-proposal pairs. The judge said "uses a claim" 10 times, the human 3; all 7 disagreements were a machine yes against a human no, and the judge never missed a true yes. In four, the proposal cites a result from another report; three are judgment calls. The judge trusts a proposal's own account of where its idea came from, so its 62% uptake rate overstates communication between islands (outputs/human_review_submission.md, Opus study).

Reflection. Claude was good at fast, wide first-pass audits (P1 listed every place the mapping is approximate), at structured scepticism (P4 proposed the standard battery) and at following a blind protocol (98 of 108 valid proposals; blinding audit passed). It was bad at calibration (design choices became "confirmed errors"; a p -value it never computed), at holding rules fixed (new thresholds after seeing results), at counting and formats, at judging provenance (7 false yes in 17) and at seeing past its feedback: shown IC by tightness tercile, the agents built regime gates. Sharing did not help. Claude is a useful auditor and idea generator when a human checks every claim against the code and data, and when it cannot change the rules after seeing results.

TABLE 7. Claude as research assistant: six interactions (what we asked, in the first column; prompts in Section 2.3, detail in Appendix A). CONFIRMATORY (P1–P4) / POST-HOC (P5–P6)

Interaction	What it got right	What it got wrong	What we changed	Our critique
P1 audit (Sonnet): find mapping, look-ahead and code bugs in the manifest, crosswalk and as-of code	Flagged every real scope gap: transport without utilities, Telcm for Information, groups 9 and 10 sharing one CES series	Called design choices “confirmed errors”; 5 of 25 findings wrong (the “vanished” group-10 series had 584 shared rows); no look-ahead bug	All 25 tested; one metadata fix to the crosswalk; no number or timing rule changed	[TODO: critique: Alex]
P2 agents (Sonnet; 3 independent, 3 islands, 3 seeds): blind to names and dates, propose six features each in a 14-operation grammar	98 of 108 proposals valid; blinding audit of all 199 calls passed; best research IC 0.08 independent, 0.05 islands	Both independent-arm picks switch on only in the low-tightness tercile; 11 of 18 migration reports broke the rules, 10 over 120 words	Invalid proposals used their slot; selection mechanical, two per arm, seed 1 only; no engineer edits	IC shown by tightness tercile pushed agents to regime gates; both A3 picks were off for most of the test
P3 judge (Sonnet; Opus in the follow-up): does a later island proposal use or test a claim from an earlier report?	Follow-up: agreed with the human on 10 of 17 pairs; caught all 3 human “yes” cases	Main run: 28 of 36 answers not clean JSON, 8 usable; follow-up: all 7 disagreements false “yes”	Undefined labels kept as missing; no repair parsing or extra calls; uptake reported as descriptive	All 7 disagreements were false positives, not a random mix – a judge that scores on topic/vocabulary overlap will drift toward “yes” wherever reports share wording. With only 17 audited pairs we can’t rule out noise, but the one-sided error pattern points to a concrete fix: require the judge to quote the exact borrowed claim, not just flag a shared topic.
P4 red team (Sonnet): from anonymous research tables, ten reasons the results could be spurious, each with a test	Read the tables correctly: no IC t above 1.3, negative Sharpes; 6 of 10 concerns matched registered tests	Stated a permutation p it never computed; proposed thresholds after seeing results (95th percentile, DSR 0.95, rolling 24 months)	Registered diagnostics kept; the 4 new tests and every new threshold rejected as post hoc	[TODO: critique: Romain]
P5 post-mortem (Opus, after the sealed test): why did the frozen strategies lose money?	Agent features active in about 3 of 142 test months, which fits the low- T gate: T stayed high after 2014	Its fixes (wage growth back, two A0 signs flipped) read the sealed results; its “stable” wage signal: -1.71 over the last 18 months	v2: four variants frozen before running, all reported, W the headline; labelled post hoc (docs/FINDINGS_AND_PROPOSAL.md)	[TODO: critique: Piero]
P6 iteration (Opus, Fable): rebuild the analysis, freeze and run v2/v3, build the forward book, draft this report	Reproduced the sealed returns to 10^{-16} ; froze every spec before running; stopped at a failed gate	Assumed August JOLTS would not be out by 29 September; it was, so its own check stopped the run	Amendment 001, disclosed; run logs keep every result reproducible	[TODO: critique: Alex]

References

- David H. Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.
- Frederico Belo, Xiaoji Lin, and Santiago Bazdresch. Labor hiring, investment, and stock return predictability in the cross section. *Journal of Political Economy*, 122(1):129–177, 2014.
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- Carlo Zarattini and Gary Antonacci. A century of profitable industry trends. SSRN Working Paper 4857230, 2024.

A Claude, interaction by interaction

Condensed from the team's written evaluation (Al Yazid Bensaid, 3 October 2026). Paths are relative to `study1_preregistered/outputs/follow_the_workers/` unless stated; the Opus follow-up has the same files under `follow_the_workers_opus55_xhigh/`.

Claude took every AI role: Sonnet 5.5 for the confirmatory study, Opus 5.5 at extra-high effort for the follow-up. This term's course allows it; the switch from the ChatGPT template is logged as preregistration amendments 001 and 002 (`outputs/prereg_amendments/`, `outputs/deviations.md`), and the log file keeps the template name `chatgpt_log.md` with Claude content.

P1, data and code audit. The system prompt made Claude “a careful data and code auditor” with no tools; the prompt in Section 2.3 was followed by the manifest, the crosswalk and the as-of code (38,717 characters). It returned 25 numbered findings, all tested (`outputs/p1_review.md`): five were claims the files did not support, among them that group 10's CES series “silently vanishes” (584 shared, nonmissing rows were present); one led to a metadata fix (the crosswalk now separates the prescribed match from the verified French scope); the rest were documented design choices or limitations, and none was a look-ahead error.

P2, signal agents. Six agents (three independent, three islands) over three seeds proposed six features each in a 14-operation grammar, blind to names and dates, with research-period feedback only. 98 of 108 proposals passed validation; the blinding audit of all 199 calls passed (108 proposals, 18 migration reports, one syntax repair). Islands exchanged migration reports capped at 120 words; 11 of 18 were rejected, 10 of them for running to 121–133 words although the prompt asked the agent to check its count. Selection was mechanical, two per arm from seed 1 only, and no engineer edited a proposal. Both independent-arm picks, decoded after the test, were low-tightness gates: `regime(yoy(Earnings), "low")` and `regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")`, active in 3 of 142 test months. The agents optimised what they were shown: IC split by tightness tercile. Blinded, they could not know that T would stay high after 2014. Next time, keep regime splits out of agent feedback, or require a minimum share of active months in research.

P3, judge. In the follow-up, items 2, 3 and 13 cite “submission 2's significant negative IC”, a result that exists in another report; item 12 cites “submission 4” from a report with two submissions. Items 8, 16 and 17 are judgment calls (a methodological tip reused; a report's topic shared but its result contradicted). The same proposal paired with two reports (items 12 and 17) got a yes both times, which is what a judge that matches on vocabulary would do. With 17 dependent pairs the sample is small. In the confirmatory study only 8 of 36 judge outputs were usable and 8 pairs were audited.

P4, red team. Six of the ten concerns were already covered by registered diagnostics (placebo, 25 bp costs, test halves, tightness, data-vintage variants); for the other four no additional test was adopted. The response asserted a permutation p -value “already” above 0.10 that nobody had computed, and proposed a 95th-percentile gate, a DSR of 0.95 and a rolling 24-month threshold after seeing the research tables.

P5, post-mortem. Written from the sealed results, so its fixes are post hoc by construction; the review's own errors (a vendor guess from a code path, a momentum definition that included the latest month, wage ICs from a non-frozen start date) are recorded with their corrections in `docs/FINDINGS_AND_PROPOSAL.md`.

Bottom line. Every Claude output needed a human check: 25 P1 dispositions, 10 P4 dispositions and 25 hand-labelled migration pairs (8 Sonnet, 17 Opus).

A.1 Transcripts and verbatim excerpts

- P1: prompt `outputs/p1_request.json` (the instruction in Section 2.3 followed by the manifest, crosswalk and as-of code); response `outputs/p1_response.md`; dispositions `outputs/p1_review.md`: 25 findings, 5 not supported by the files, 1 metadata fix, 19 documented design choices, limitations or checks that found no error.
- P2: system and turn prompts in `params.py` (`SYSTEM_PROMPT`, `TURN_PROMPT`); every call with its raw response in `outputs/agents/log.jsonl`; proposals in `outputs/candidates.csv`; selected features decoded in `analysis/output/study1/tables/d7_selected_features.csv`.
- P3: migration reports in `outputs/agents/migrations.json`; the judge prompt in `agents.py`, function `trace_migrations`; judgments in `outputs/agents/migration_judgments.json`; the human audit in `outputs/human_review_submission.json` (the Opus folder also has a `.md` version).
- P4: prompt `outputs/p4_request.json`; response `outputs/p4_response.md`; dispositions `outputs/p4_review.md`.
- P5: the review and its output, `docs/FINDINGS_AND_PROPOSAL.md`, with the numbers in `analysis/output/study1/diagnostics.md`.
- P6: three task prompts (rebuild and diagnostics; v2/v3 freeze, forward test and report draft; final repository pass) and a one-page rules file (frozen folder read-only, one label per result, specs hashed before they run, a source comment on every number, stop at a failed gate), kept outside the repository; outputs in `analysis/run.ipynb`, `analysis/output/`, `results/` and this report.

Verbatim excerpts. Four short excerpts, copied from the files above without editing.

P1 OUTPUT, OPENING OF THE RESPONSE (`OUTPUTS/P1_RESPONSE.MD`)

Scope: only the supplied material. I did not verify live FRED data. I mark each item as confirmed (visible in the text) or as a question or limitation. Confirmed errors: mapping and manifest. 1. Manifest rows for CEU5500000007 and CEU5500000008 with group 10 duplicate the group 9 rows. The same series ID is mapped to two groups, and group 10 (Real estate) has no distinct CES series. [...] Test: group the manifest by id and assert one group per series. Compare the feature columns of groups 9 and 10, which will be identical.

P2 OUTPUT, ONE AGENT PROPOSAL, DECODED AFTER THE TEST (INDEPENDENT ARM, REPETITION 1, TEMPORAL AGENT, PROPOSAL 3)

Expression regime(yoy(V05),"low"). Hypothesis: groups with higher year-over-year change in V05 (dollars per hour) will have higher next-month returns in low-T regimes, because accelerating compensation signals stronger conditions not yet fully priced in. The effect should vanish in high T. [V05 was later decoded as average hourly earnings; the feature was selected for A3 and was active in 3 of 142 test months.]

P3 JUDGE VERSUS HUMAN, TWO AUDITED PAIRS (`OUTPUTS/HUMAN_REVIEW_SUBMISSION.MD`, OPUS STUDY)

Item 5, human yes, judge yes: "The report found that V02 outpacing V03 led to significant underperformance, and that the V03-favoring reversal was positive across T regimes. The proposal uses that V03/V02 composition signal without conditioning on T." Item 2, human no, judge yes: "It says it reverses 'an observed negative signal,' but this report has no negative signal: the V02/V03 ratio had IC 0.01 and the V02-vs-V04 shift had IC 0."

P4 OUTPUT, FIRST OF TEN REASONS (`OUTPUTS/P4_RESPONSE.MD`)

1. The IC signals are pure noise (no predictive content). All t-statistics are below 1.3 and the signs flip across variants. Test: a block-bootstrap or permutation test of IC, shuffling

signals within each date, 1000 or more draws, with Newey-West lags of 6. Do not implement if: the permutation p-value is above 0.10 for any variant. This is already the case, so the answer is do not implement. [Disposition in p4_review.md: covered by the registered circular-shift placebo; no new test adopted.]

B Data sources and links

- Ken French Data Library: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html (49 industry portfolios, five factors, momentum; CRSP build 202608). Live, checked 3 October 2026.
- JOLTS by industry, not seasonally adjusted: FRED series JTU<code>JOR/HIR/QUR/LDR, for example <https://fred.stlouisfed.org/series/JTU2300HIR> (checked 3 October 2026, series runs through Aug 2026). Codes 110099, 2300, 3200, 3400, 4200, 4400, 480099, 5100, 5200, 5300, 540099, 6200, 7200.
- CES hours and earnings: CEU<code>07 and CEU<code>08 (construction earnings CES2000000008); codes 10000000, 20000000, 31000000, 32000000, 41420000, 42000000, 43000000, 50000000, 55000000, 60000000, 65000000, 70000000. Checked 3 October 2026 via FRED.
- Tightness: JTSJOL and UNEMPLOY. Vintages: <https://alfred.stlouisfed.org/> (live, checked 3 October 2026).
- ETF proxies and their caveats: [analysis/output/forward/etf_proxies.csv](#).

C From 49 French industries to 13 labor groups

TABLE 8. The 13 groups: JOLTS code, Ken French members and the largest member's share of August 2026 market capitalisation. CONFIRMATORY

Group	JOLTS	Ken French industries	Largest member
Mining and logging	110099	Gold, Mines, Coal, Oil	Oil 79%
Construction	2300	Cnstr	100%
Durable manufacturing	3200	Toys, BldMt, Steel, FabPr, Mach, ElcEq, Autos, Aero, Ships, Guns, Hardw, Chips, LabEq, MedEq	Chips 60%
Nondurable manufacturing	3400	Food, Soda, Beer, Smoke, Hshld, Clths, Drugs, Chems, Rubbr, Txtls, Paper	Drugs 60%
Wholesale	4200	Whsl	100%
Retail	4400	Rtail	100%
Transport and utilities	480099	Trans, Util	Util 61%
Information	5100	Telecm	100%
Finance and insurance	5200	Banks, Insur, Fin	Banks 51%
Real estate	5300	REst	100%
Professional and business services	540099	BusSv	100%
Health care	6200	Hlth	100%
Accommodation and food	7200	Meals	100%

Groups 9 and 10 share one CES series, and groups 7, 12 and 13 use broader CES supersectors, so the effective cross-sectional width is closer to 8 or 9 than 13.

D Additional results and figures

Data vintages. A3 test Sharpe: as-known -0.58 , first release -0.52 , revised -0.56 , fixed lags -0.53 ; IC -0.021 , -0.040 , -0.026 , -0.021 .

Cost sensitivity. A3 Sharpe -0.45 at 0 bp and -0.79 at 25 bp industry costs. HQ12: $+0.34$, $+0.29$, $+0.22$ and $+0.09$ at 0, 10, 25 and 50 bp; break-even 68 bp.

Gates. G1 timing: pass. G2 prediction ($IC\ t \geq 2$): fail. G3 incremental alpha ($t \geq 2$): fail. G4 robustness (positive IC in both halves, placebo $p < 0.05$, no industry above half the profit): fail. G5 implementable (Sharpe > 0 at 25 bp, capacity documented): fail.

Primary by window. First half -0.72 , second half -0.49 , excluding March–December 2020 -0.51 , last 18 months $+0.85$.

Where the numbers live. Signal, horizon, tightness, risk-overlay and agent tables: `analysis/output/study1/diagnostics_v2.md`; v2 statistics: `analysis/output/study2/v2_results.md`; the robustness battery: `analysis/output/study2/v3_results.md`; every claim with its source: `results/claims.csv`.

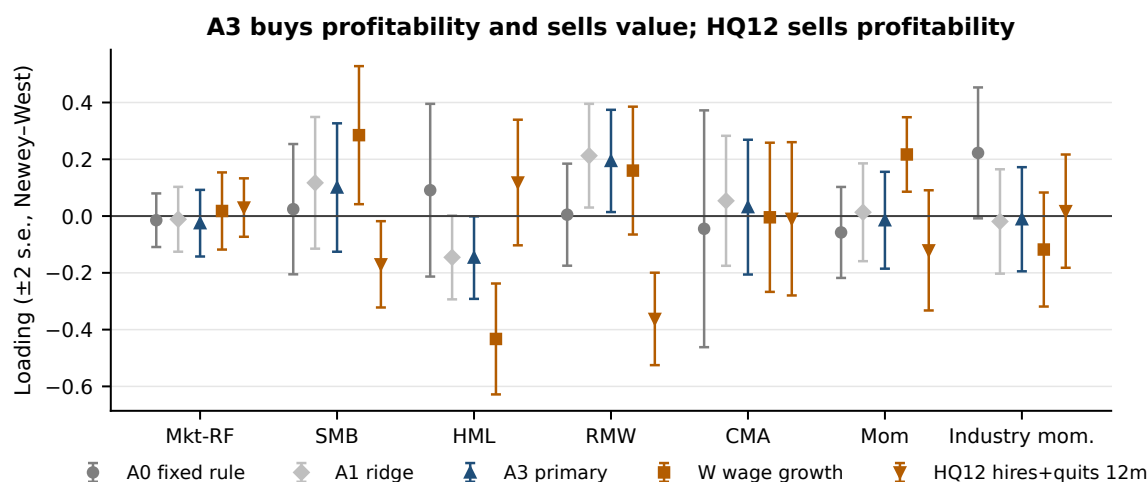


FIGURE 6. Factor loadings of the long-short books: A3 buys profitability and sells value; W sells value and buys momentum; HQ12 sells profitability. CONFIRMATORY (A0, A1, A3) / POST-HOC (W, HQ12)

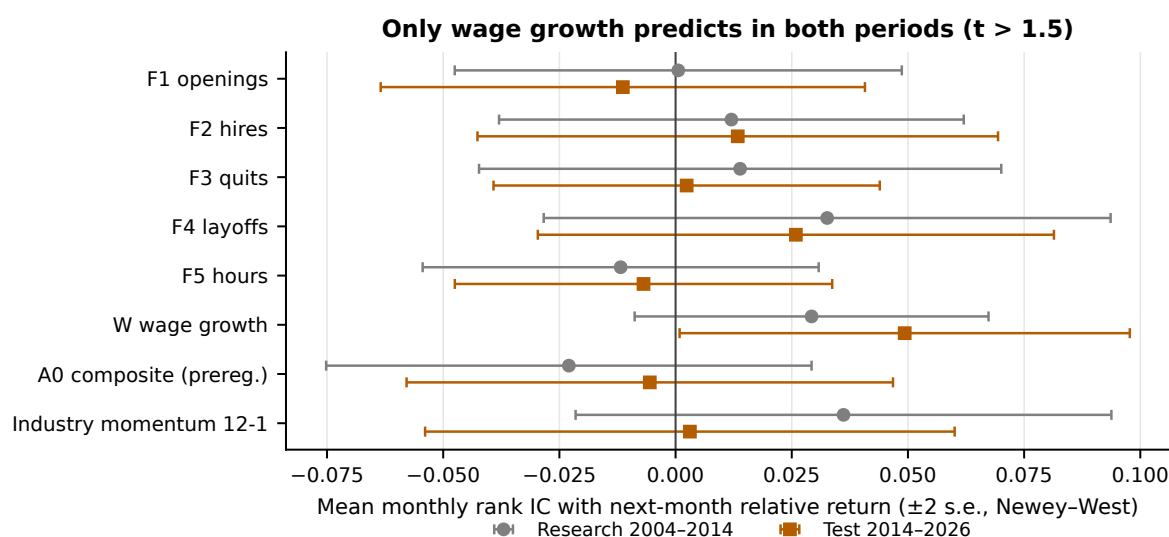


FIGURE 7. Only wage growth has an IC above 1.5 standard errors in both periods; layoffs and hours carry the opposite sign to the one the fixed rule imposed. POST-HOC

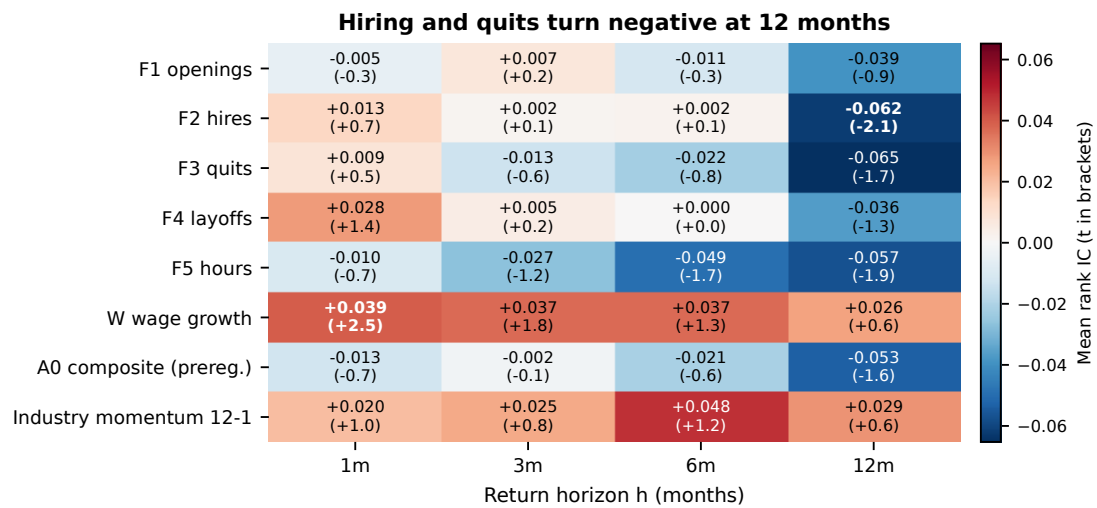


FIGURE 8. Hires and quits turn negative at 12 months ($t = -2.1, -1.7$): the investment channel shows up slowly, not within a month. POST-HOC

TABLE 9. Net Sharpe ratio by window.

CONFIRMATORY (A3) / POST-HOC (W, HQ12)

Window (returns)	Research 2004–14	Test 2014–26	Full sample 2004–26	Post-2010 2010–26	Last 18 months 2025-03–2026-08
A3 primary, H1 (confirmatory)	−0.13	−0.58	−0.44	−0.62	+0.85
W wage growth (post-hoc)	+0.09	+0.16	+0.15	+0.12	−1.71
HQ12 −(hires+quits), 12m, H2 (post-hoc)	+0.50	+0.29	+0.35	+0.30	+0.29

A3's research figure uses the 88 months with a forecast; its full-sample figure starts in June 2007 and is descriptive. W and HQ12 use revised data with fixed lags over 125, 142, 268, 200 and 18 months.

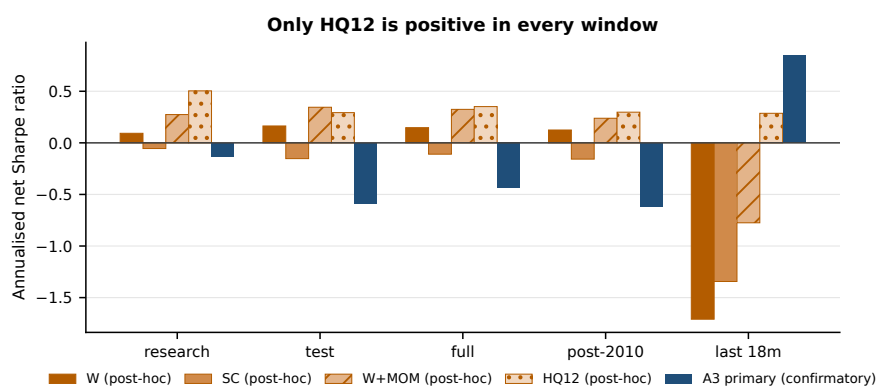


FIGURE 9. Net Sharpe by window. HQ12 is the only post-hoc variant positive in every window; W and W+MOM lost heavily in the last 18 months. POST-HOC / CONFIRMATORY (A3 bars)

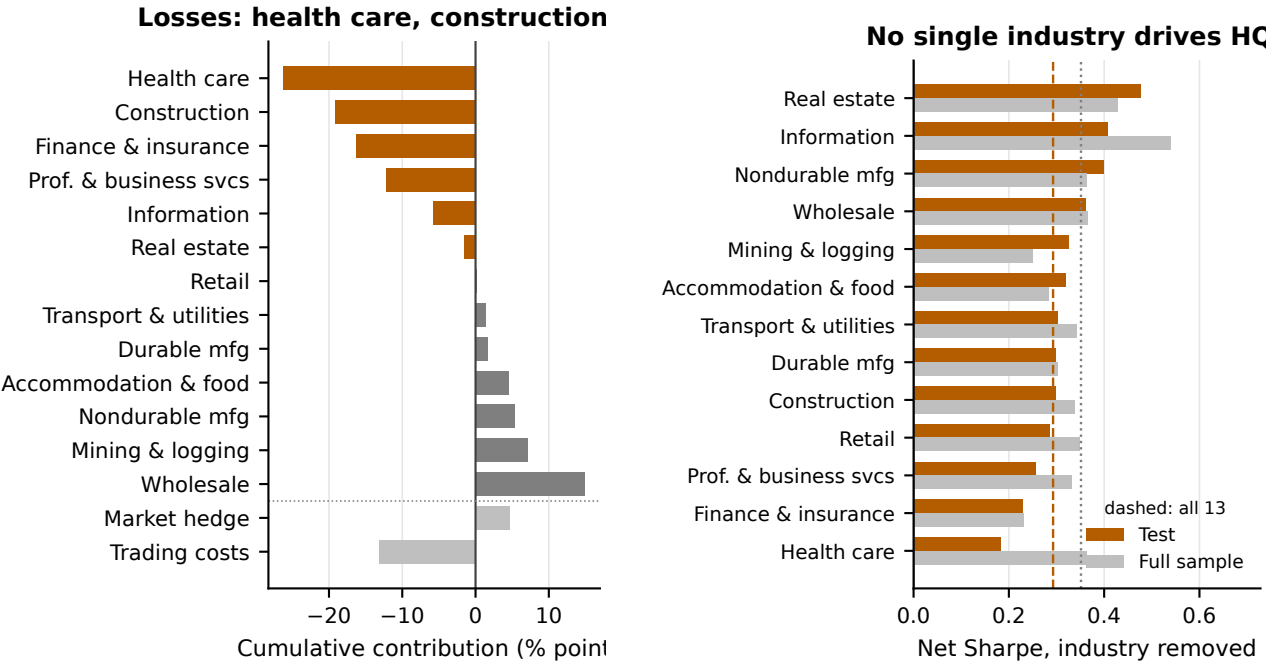


FIGURE 10. Left: the primary lost most in health care, construction and finance. Right: no single industry carries HQ12; removing any one leaves its test Sharpe between +0.18 and +0.48.

POST-HOC

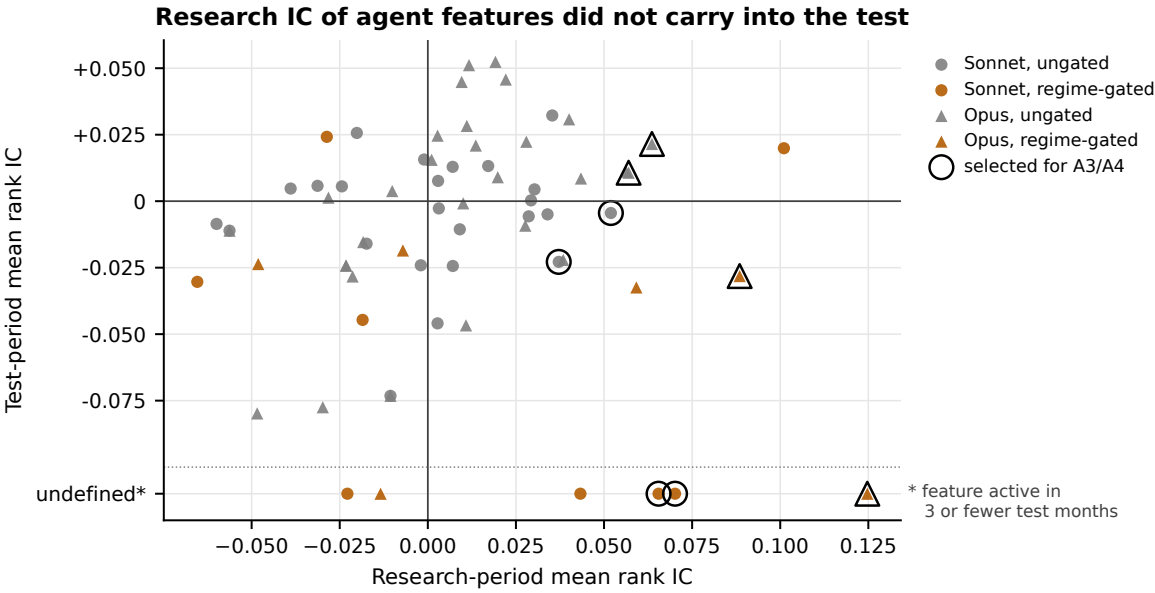


FIGURE 11. Research IC of agent features barely predicts their test IC; the selected regime-gated features were active in 3 test months.

POST-HOC

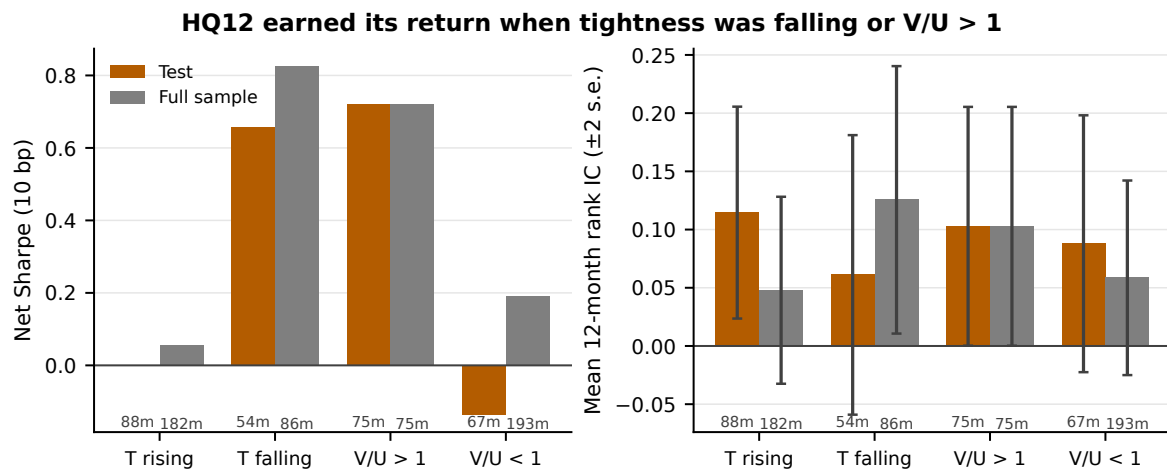


FIGURE 12. HQ12 conditional on tightness at the decision, recorded without a claim: the return came in months when tightness was falling or $V/U > 1$, while the 12-month IC was highest when T was rising (+0.115, $t = 2.52$); 54 to 88 months per cell. POST-HOC

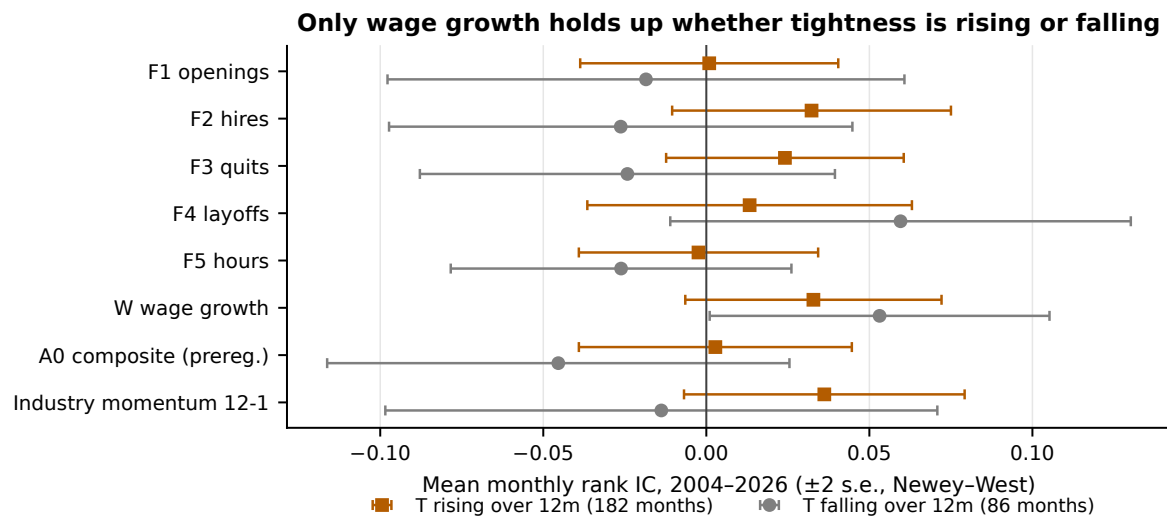


FIGURE 13. Single-signal IC in months when T rose versus fell over the previous 12 months: only wage growth keeps an IC above 1.5 standard errors in both states; the split is balanced in both periods. POST-HOC

TABLE 10. Rank IC by horizon, May 2004–August 2026 (t in brackets). POST-HOC

Signal	1m	3m	6m	12m
F1 openings	−0.005 (−0.31)	+0.007 (+0.23)	−0.011 (−0.28)	−0.039 (−0.92)
F2 hires	+0.013 (+0.70)	+0.002 (+0.07)	+0.002 (+0.08)	−0.062 (−2.08)
F3 quits	+0.009 (+0.50)	−0.013 (−0.55)	−0.022 (−0.78)	−0.065 (−1.73)
F4 layoffs	+0.028 (+1.37)	+0.005 (+0.17)	+0.000 (+0.01)	−0.036 (−1.28)
F5 hours	−0.010 (−0.67)	−0.027 (−1.24)	−0.049 (−1.69)	−0.057 (−1.86)
W wage growth	+0.039 (+2.51)	+0.037 (+1.77)	+0.037 (+1.28)	+0.026 (+0.63)
A0 composite (prereg.)	−0.013 (−0.69)	−0.002 (−0.09)	−0.021 (−0.61)	−0.053 (−1.58)
Industry momentum 12-1	+0.020 (+0.99)	+0.025 (+0.80)	+0.048 (+1.24)	+0.029 (+0.63)

TABLE 11. HQ12 robustness battery: holding period (R3) and sub-periods (R4).

POST-HOC

Holding	research	test	full	post-2010	last 18m	Test turnover
6 months	+0.31	+0.05	+0.15	+0.05	+0.50	0.49
9 months	+0.43	+0.14	+0.24	+0.16	+0.29	0.40
12 months	+0.50	+0.29	+0.35	+0.30	+0.29	0.37
18 months	+0.58	+0.30	+0.38	+0.34	+0.31	0.28

Window	Months	Sharpe	Net p.a.	IC 12m (t)
test: first half	71	-0.21	-1.4%	+0.097 (+1.75)
test: second half	71	+0.64	+6.5%	+0.095 (+1.60)
test: excl. Mar 2020-Dec 2021	120	+0.37	+3.0%	+0.105 (+2.23)
test: pre-2020	62	+0.14	+0.8%	+0.114 (+1.92)
test: 2020 onward	80	+0.37	+3.9%	+0.080 (+1.54)
full: first half	134	+0.33	+2.0%	+0.038 (+0.73)
full: second half	134	+0.38	+3.3%	+0.107 (+2.66)
full: excl. Mar 2020-Dec 2021	246	+0.40	+2.9%	+0.073 (+1.93)
full: pre-2020	188	+0.36	+2.2%	+0.068 (+1.57)

Positive at every holding period, 6 to 18 m

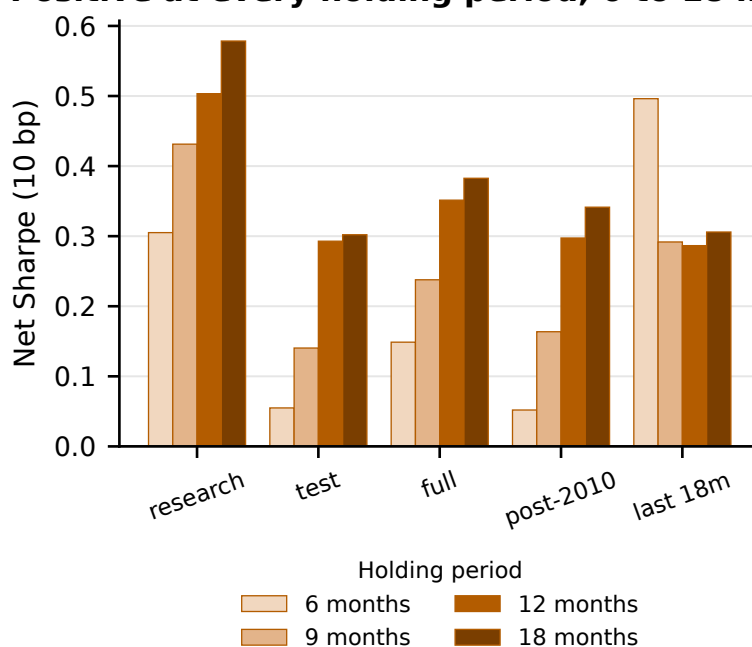


FIGURE 14. HQ12 is positive at every holding period from 6 to 18 months (test Sharpe +0.14, +0.29, +0.30 at 9, 12 and 18).

POST-HOC

TABLE 12. HQ12 drop-one-industry results (R7), test window: net Sharpe with each group removed. **POST-HOC**

industry	dropped_group	test_sharpe	full_sharpe
Mining & logging	1	0.3254132438206908	0.25091135502654416
Construction	2	0.2987158360392865	0.3389966713324877
Durable mfg	3	0.29903781053527545	0.3035811529738324
Nondurable mfg	4	0.39895252195406133	0.36450850554728964
Wholesale	5	0.36083932009982755	0.3650139867674817
Retail	6	0.2865654382978325	0.3485857233046736
Transport & utilities	7	0.3026891231778837	0.34344805121872624
Information	8	0.4080462959220044	0.5393726246166338
Finance & insurance	9	0.22962454708886865	0.2313849406616241
Real estate	10	0.4765507360777538	0.4289078032675492
Prof. & business svcs	11	0.25688387453327266	0.3333615491811263
Health care	12	0.18405490659750104	0.36319371802310363
Accommodation & food	13	0.31910398743048873	0.2846190379118023

E Key code

As-of lookup. The value of a series known on a cutoff date (frozen `data.py`).

```
def vintage_view(frame, cutoff, first=False):
    cutoff = pd.Timestamp(cutoff).strftime('%Y-%m-%d')
    if first:
        eligible = frame.drop_duplicates('observation_month', keep='first')
        eligible = eligible.loc[eligible.realtime_start <= cutoff]
    else:
        eligible = frame.loc[(frame.realtime_start <= cutoff) & (frame.realtime_end >= cutoff)]
    if eligible.duplicated('observation_month').any():
        raise ValueError('Overlapping vintage intervals')
    return eligible.set_index('observation_month')[['value', 'realtime_start']].sort_index()
```

Signal construction. Fixed-lag signals and tightness (`analysis/data.py`).

```
def signals(folder=P.FRED, last_decision=None, groups=None):
    panel = labor_panel(folder)
    avg3 = lambda x: x.rolling(3, min_periods=3).mean()
    raw = {}
    to_decision = lambda frame, lag: frame.set_axis(frame.index + lag) # observation o ->
    decision o+lag
    for i, m in enumerate(P.JOLTS_MEASURES, 1):
        x = avg3(panel[m])
        raw[f'F{i}'] = to_decision(x - x.shift(12), P.JOLTS_LAG)
    h, e = avg3(panel['H']), avg3(panel['E'])
    raw['F5'] = to_decision(np.log(h / h.shift(12)), P.CES_LAG)
    raw['W'] = to_decision(np.log(e / e.shift(12)), P.CES_LAG)
    # last decision month at which every feature has its lagged observation
    end = last_decision or min(v.dropna(how='all').index.max() for v in raw.values())
    decisions = pd.period_range(P.SIGNAL_START, end, freq='M')
    features = {}
    for name, value in raw.items():
        value = value[list(groups)] if groups is not None else value # drop-one-industry
        # diagnostics
        z = standardize(value.reindex(decisions))
        z.index = z.index + 1 # earning month
        features[name] = z
    # T at decision d uses the latest month <= d-2 with both V and U observed (frozen rule); this
    # matters once: the Oct 2025 household survey (UNEMPLOY) was never collected.
    ratio = np.log(panel['V'] / panel['U']).ffill()
    t = to_decision(ratio, P.JOLTS_LAG).reindex(decisions)
    t.index = t.index + 1
```

```
return features, t
```

Backtest core. Tranche, hedge, scaling and costs (frozen `data.py`, excerpt).

```
for month,score in scores.iterrows():
    tranche = pd.Series(0.,index=groups)
    if signal_available.loc[month]:
        rank = score.sort_index().sort_values(kind='stable')
        tranche.loc[rank.index[-P.PORTFOLIO['longs']:]] = 1/P.PORTFOLIO['longs']
        tranche.loc[rank.index[:P.PORTFOLIO['shorts']]] = -1/P.PORTFOLIO['shorts']
    tranches.append(tranche)
    book = sum(tranches[-holding:])/holding
    betas = pd.Series({g:np.cov(historical[g],market,ddof=1)[0,1]/variance for g in groups})
    raw = np.r_[book.to_numpy(),-float(book@betas)]
    covariance = LedoitWolf().fit(historical.to_numpy()).covariance_
    risk = np.sqrt(max(0,float(raw@covariance@raw))*12)
    if risk<=0 and np.abs(raw).sum(>1e-14:
        raise ValueError('Degenerate portfolio risk estimate')
    scale = min(P.PORTFOLIO['vol_target']/risk,P.PORTFOLIO['gross_cap']/np.abs(raw).sum()) if
risk>0 else 0.
    weights = pd.Series(raw*scale,index=pretrade.index)
    assert abs(weights[groups]@betas+weights['market'])<1e-10
    assert weights.abs().sum()<=P.PORTFOLIO['gross_cap']+1e-10
    turnover = float((weights[groups]-pretrade[groups]).abs().sum())
    hedge_turnover = abs(weights['market']-pretrade['market'])
    transaction = turnover*cost_bps/10000+hedge_turnover*P.PORTFOLIO['hedge_cost_bps']/10000
    # Charge borrowed securities plus negative residual cash, at the stated annual rate.
    borrow_base = -weights.clip(upper=0).sum()
    finance_base = max(0,float(weights.sum()-1))
    financing = (borrow_base+finance_base)*borrow_bps/10000/12
    gross = float(weights[groups]@returns['excess'].loc[month,groups]+weights['market']*factors
['Mkt-RF'])
    net = gross-transaction-financing
```

F Opus follow-up study (exploratory)

After the test had been opened, the agent experiment was rerun with Claude Opus 5.5 at extra-high effort, with fresh agents and the same frozen evaluator, selection rule and portfolio. Because the test period was already known to the team, the run is exploratory. Its research-selected primary, A4, earned a test Sharpe of +0.01 (research -0.40) with an IC of -0.002 ($t = -0.10$), an alpha t of 0.05, a placebo p of 0.56 and a Sharpe of -0.19 at 25 bp; it failed gates G2 to G5 and had a Sharpe of -1.41 over the last 18 months. Its gross return came from one long position: durable manufacturing contributed +0.27 of cumulative gross return (sum of monthly contributions, in units of the book) in a group that is 60% semiconductors. Opus's other strategy, A3, earned -0.46. The follow-up is evidence about agents, not about the strategy. A stronger model followed the protocol better and found no more signal: 106 of 108 valid proposals against 98, 15 of 18 admitted migration reports against 7, 84 of 88 usable judge labels against 8 of 36, yet no selected feature survived the test criterion in either study. Communication raised research search scores for Opus and lowered them for Sonnet; neither gap exceeded the spread across repetitions.

G Forward test: frozen specification and Q4 2026 books

- `analysis/specs/study2_1_variants.md`: four post-hoc variants and a reading rule; SHA-256 `ca658234...`, frozen 2026-10-03T09:00:13Z, first run 09:01:55Z.
- `analysis/specs/forward_spec.md`: decision 30 September 2026, information cutoff 29 September

ber, one tranche held to 31 December, 60-month beta hedge in SPY, 10 bp and 2 bp costs; SHA-256 735a8ffa..., frozen 09:03:56Z.

- `analysis/specs/forward_amendment_001.md`: the fixed-lag information set (JOLTS July 2026, CES August 2026, as published on 29 September) after the pre-committed check stopped the first run; W+MOM and HQ12 added after the historical v2 results were seen and before any Q4 return beyond 1–2 October; SHA-256 c76cde86..., frozen 09:16:11Z; book built 09:17:55Z.
- `analysis/specs/study2_2_hq12_robustness.md`: the robustness battery; SHA-256 312aef1d..., frozen 09:52:27Z, first run 09:53:32Z.
- Books: `analysis/output/forward/book_2026Q4.csv` (group weights for W–HQ12), `book_2026Q4_etf.csv` (netted ETF weights), `book_2026Q4_hedge.csv` (SPY hedge per unit: -0.40 , -0.06 , -0.24 , -0.02), `etf_proxies.csv`. Holdings: W is long construction, durable manufacturing and information and short mining, health care and accommodation; SC is long durable manufacturing, retail and transport and short real estate, health care and accommodation; W+MOM is long construction, durable manufacturing and finance and short retail, transport and professional services; HQ12 is long finance, real estate and accommodation and short mining, durable manufacturing and retail. W+MOM's tradable book has four legs, because its long durable-manufacturing and short professional-services positions both map to the industrials ETF and cancel.
- Power: HQ12's test volatility is about 9% a year, so one quarter's return has a standard deviation near 5%; a Q4 return between -10% and $+10\%$ is within two standard deviations of zero. The book was first built at 09:17:55Z on 3 October: commit 3e8ecd7 (09:18:29Z) holds a byte-identical `book_2026Q4.csv` (SHA-256 857275...); `book_first_build.sha256` records that first build and later notebook runs must reproduce it.
- Evaluation: the ETF books from the 30 September close to the 31 December close after costs, and the same books on Ken French group returns once published; reported whatever they show. Cadence: a weekly tracker was considered and deferred by team decision – with one quarter of data and no capital at risk, a weekly readout adds noise without adding power (Section 3.3, Limits and power). We'll evaluate once at the 31 December close.